MATH423: Introduction to Computational Topology

Mehmetcik Pamuk (mpamuk@metu.edu.tr)
Department of Mathematics, Middle East Technical University

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1 Week 1: Basic Topology

To prepare for our work in computational topology, we first establish the foundational language of general topology. Topology studies properties of spaces that remain unchanged under continuous deformations, focusing on qualitative rather than quantitative features. A coffee mug and a doughnut are the classical example: one can be deformed into the other without tearing or gluing, and both have exactly one "hole". Much of this course is about making the word "hole" precise and computable.

1.1 Topological Spaces and Metric Spaces

Definition 1.1. Let X be a set. A **topology** on X is a collection $\mathcal{T}$ of subsets of X satisfying the following properties:

1. $\emptyset$ and X are in $\mathcal{T}$.
2. The union of the elements of any subcollection of $\mathcal{T}$ is in $\mathcal{T}$.
3. The intersection of the elements of any finite subcollection of $\mathcal{T}$ is in $\mathcal{T}$.

A set X equipped with a topology $\mathcal{T}$ is a **topological space**. The sets in $\mathcal{T}$ are called **open sets**. A subset $C \subset X$ is **closed** if its complement $X \setminus C$ is open.

Taking complements in the axioms, we see that $\emptyset$ and X are closed, arbitrary intersections of closed sets are closed, and finite unions of closed sets are closed. Note that "closed" is not the opposite of "open": a set may be both, or neither.

Example 1.2. Let $X = \{a, b, c\}$.

- $\mathcal{T}_1 = \{\emptyset, X\}$ is a topology, called the **trivial** (or indiscrete) topology. It exists on every set.
- $\mathcal{T}_2 = \{\emptyset, \{a\}, \{a, b\}, X\}$ is a topology. Here $\{c\}$ and $\{b, c\}$ are closed, $\{a, b\}$ is open but not closed, and $\{b\}$ is neither open nor closed.
- $\mathcal{T}_3 = \{\emptyset, \{a\}, \{b\}, X\}$ is *not* a topology, since $\{a\} \cup \{b\} = \{a, b\} \notin \mathcal{T}_3$.
- The power set $\mathcal{P}(X)$ is a topology, called the **discrete** topology.

Example 1.3. The requirement that only *finite* intersections of open sets be open is essential. In $\mathbb{R}$ with its usual open intervals, the sets $U_n = (-\frac{1}{n}, \frac{1}{n})$ are open for every $n \geq 1$, but

$$\bigcap_{n=1}^{\infty} \left(-\tfrac{1}{n}, \tfrac{1}{n}\right) = \{0\},$$

which is not open in $\mathbb{R}$.

In applied settings, such as topological data analysis, our spaces often come from point clouds equipped with a strict notion of distance.

Definition 1.4. A **metric space** is a set X together with a distance function $d : X \times X \to \mathbb{R}$ such that for all $x, y, z \in X$:

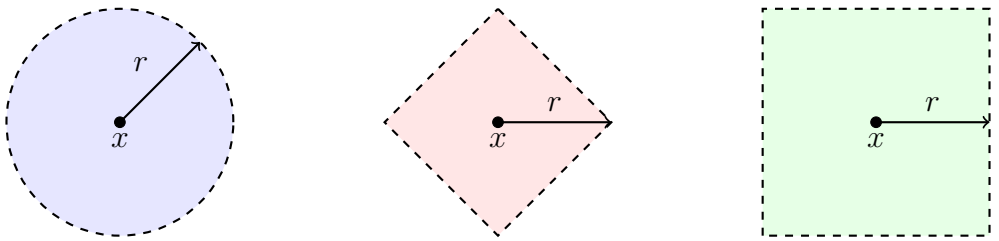
1. $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$.
2. $d(x, y) = d(y, x)$ (symmetry).
3. $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality).

Every metric space naturally induces a topology. We define the open ball of radius $r > 0$ centered at $x \in X$ as $B_r(x) = \{y \in X \mid d(x, y) < r\}$, and the closed ball as $\overline{B}_r(x) = \{y \in X \mid d(x, y) \leq r\}$. A set $U \subset X$ is open in the metric topology if for every $x \in U$, there exists some $r > 0$ such that $B_r(x) \subset U$. For Euclidean space $\mathbb{R}^n$, this forms the standard topology.

Example 1.5. Open balls are open in the metric topology. Let $y \in B_r(x)$ and put $s = r - d(x, y) > 0$. If $z \in B_s(y)$, then by the triangle inequality

$$d(x, z) \leq d(x, y) + d(y, z) < d(x, y) + s = r,$$

so $B_s(y) \subset B_r(x)$. Similarly, one checks that closed balls are closed sets.



L_2 (Euclidean) Ball L_1 (Manhattan) Ball L_∞ (Chebyshev) Ball

Figure 1: Open balls $B_r(x)$ in $\mathbb{R}^2$ depend on the chosen metric, though all three induce the same standard topology.

Remark 1.6. In $\mathbb{R}^n$, the most common distance functions used in data analysis are the Euclidean, Manhattan and Chebyshev metrics. Let $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$ be points in $\mathbb{R}^n$.

- **Euclidean Metric (L_2):** The standard straight-line distance, defined as

$$d_2(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

- **Manhattan Metric (L_1):** The distance measured strictly along axes at right angles, defined as
$$d_1(x, y) = \sum_{i=1}^{n} |x_i - y_i|$$

- **Chebyshev Metric (L_∞):** The largest coordinate difference,
$$d_\infty(x, y) = \max_{1 \le i \le n} |x_i - y_i|$$

While they produce geometrically different open balls, they are **topologically equivalent**; that is, they induce the exact same collection of open sets (the standard topology) on $\mathbb{R}^n$.

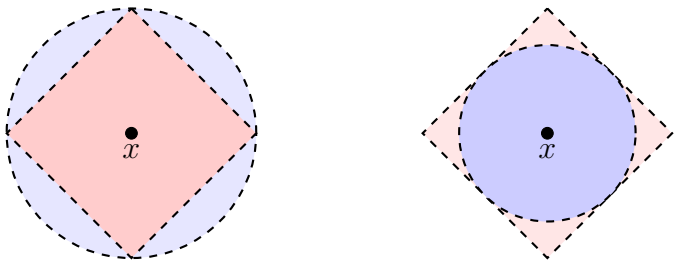
To see why for d_1 and d_2, recall that two metrics d and d' on a set X induce the same topology if and only if for every point $x \in X$ and every radius $r > 0$, there exist $r_1, r_2 > 0$ such that:
$$B_{r_1}^{(d)}(x) \subseteq B_r^{(d')}(x) \quad \text{and} \quad B_{r_2}^{(d')}(x) \subseteq B_r^{(d)}(x)$$
Geometrically, this means we can always fit a Manhattan diamond inside a Euclidean circle, and a Euclidean circle inside a Manhattan diamond. Algebraically, this follows from the standard norm inequalities in $\mathbb{R}^n$ (the second one via the Cauchy-Schwarz inequality):
$$d_2(x, y) \le d_1(x, y) \le \sqrt{n}\, d_2(x, y)$$

Consequently:

1. If $U \subseteq \mathbb{R}^n$ is open in the L_2 topology, then for any $x \in U$ there is some $r > 0$ such that $B_r^{(d_2)}(x) \subseteq U$. Since $d_2(x, y) \le d_1(x, y)$, any point y satisfying $d_1(x, y) < r$ also satisfies $d_2(x, y) < r$. Thus, $B_r^{(d_1)}(x) \subseteq B_r^{(d_2)}(x) \subseteq U$, proving U is open in the L_1 topology.

2. Conversely, if $U \subseteq \mathbb{R}^n$ is open in the L_1 topology, then for any $x \in U$ there is some $r > 0$ such that $B_r^{(d_1)}(x) \subseteq U$. The inequality $d_1(x, y) \le \sqrt{n}\, d_2(x, y)$ guarantees that $B_{r/\sqrt{n}}^{(d_2)}(x) \subseteq B_r^{(d_1)}(x) \subseteq U$, proving U is open in the L_2 topology.

The same argument with the inequalities $d_\infty(x, y) \le d_2(x, y) \le \sqrt{n}\, d_\infty(x, y)$ shows that d_∞ also induces the standard topology.



$B_r^{(d_1)}(x) \subseteq B_r^{(d_2)}(x)$ $\qquad$ $B_{r/\sqrt{2}}^{(d_2)}(x) \subseteq B_r^{(d_1)}(x)$

Figure 2: Nesting of L_1 and L_2 open balls in $\mathbb{R}^2$, illustrating that both metrics generate the exact same open sets.

Example 1.7. Let us compute the three distances between $x = (1, 2)$ and $y = (4, 6)$ in $\mathbb{R}^2$. The coordinate differences are 3 and 4, so

$$d_2(x, y) = \sqrt{9 + 16} = 5, \qquad d_1(x, y) = 3 + 4 = 7, \qquad d_\infty(x, y) = \max\{3, 4\} = 4.$$

Observe that $d_\infty \le d_2 \le d_1 \le \sqrt{2}\, d_2$, as predicted: $4 \le 5 \le 7 \le 7.07$. In data analysis the choice of metric changes the actual numbers (and hence the scales at which features appear) but, by the remark above, never the underlying topology of $\mathbb{R}^n$.

Example 1.8. Not all metrics behave like Euclidean distance. On any arbitrary set X, we can define the **discrete metric** as:

$$d(x, y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{if } x \neq y. \end{cases}$$

It is straightforward to verify that d satisfies the metric axioms (non-negativity, symmetry, and the triangle inequality). Let us examine the open balls in (X, d) for different radii $r > 0$ centered at a point $x \in X$:

- **Case 1 ($0 < r \le 1$):** By definition, $B_r(x) = \{y \in X \mid d(x, y) < r\}$. Since the distance to any other point $y \neq x$ is 1, no other point satisfies $d(x, y) < r$. For instance, choosing $r = 0.5$, we have:

  $$B_{0.5}(x) = \{y \in X \mid d(x, y) < 0.5\} = \{x\}$$

  Thus, every singleton set $\{x\}$ is an open ball, meaning every single-point subset is an open set in the induced topology.

- **Case 2 ($r > 1$):** Since the distance between any two points is at most 1, every point in X satisfies $d(x, y) < r$. Therefore, $B_r(x) = X$.

Because every singleton $\{x\}$ is open, we can use the second axiom of topological spaces (arbitrary unions of open sets are open) to see that *any* subset $A \subseteq X$ is open, as it can be written as the union of its elements:

$$A = \bigcup_{a \in A} \{a\}$$

Consequently, the topology induced by the discrete metric is the power set $\mathcal{T} = \mathcal{P}(X)$, which is called the **discrete topology**. Notice also that because the complement of every subset is open, every subset $A \subseteq X$ is simultaneously **open and closed** (clopen). Geometrically, the space consists of completely isolated points with no continuous paths connecting distinct points.

Remark 1.9. Not every topology comes from a metric. In a metric space any two distinct points $x \neq y$ can be separated by the disjoint open balls $B_r(x)$ and $B_r(y)$ with $r = d(x, y)/2$. The trivial topology on a set with at least two points has no such separating open sets, so it is not induced by any metric. This separation property deserves a name.

Definition 1.10. A topological space X is **Hausdorff** if for every pair of distinct points $x, y \in X$ there exist open sets $U \ni x$ and $V \ni y$ with $U \cap V = \emptyset$.

Every metric space is Hausdorff, and so is every subspace of $\mathbb{R}^n$. In a Hausdorff space, limits of sequences are unique, which is one reason the condition appears in the definition of a manifold in Week 2.

1.2 Subspaces and Quotient Spaces

We frequently construct new topological spaces from existing ones. The two most common methods are taking subsets (subspaces) and gluing parts together (quotients).

Definition 1.11. Let $(X, \mathcal{T})$ be a topological space and let $A \subset X$. The **subspace topology** on A is defined by $\mathcal{T}_A = \{U \cap A \mid U \in \mathcal{T}\}$.

Example 1.12. Let $A = [0, 1] \subset \mathbb{R}$. The set $[0, \frac{1}{2})$ is *not* open in $\mathbb{R}$, but it is open in A, because

$$[0, \tfrac{1}{2}) = (-1, \tfrac{1}{2}) \cap [0, 1].$$

Thus whether a set is open depends on the ambient space. Similarly, the circle $S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2+y^2 = 1\}$ inherits its topology from $\mathbb{R}^2$: its open sets are the intersections of open subsets of the plane with the circle, for instance open arcs.

The quotient topology allows us to formalize the idea of "gluing". Let X be a topological space and $\sim$ be an equivalence relation on X. The quotient space $X/\sim$ is the set of equivalence classes.

Definition 1.13. Let $p : X \to X/\sim$ be the natural projection map sending each element to its equivalence class. The **quotient topology** on $X/\sim$ is defined by declaring a subset $U \subset X/\sim$ to be open if and only if its preimage $p^{-1}(U)$ is open in X.

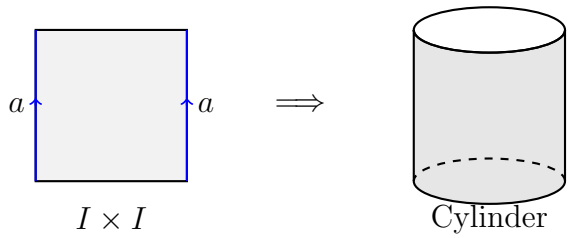


Figure 3: Forming a quotient space. Identifying the left and right edges of the unit square $I \times I$ according to the relation $(0, y) \sim (1, y)$ produces a cylinder.

Example 1.14. Consider the interval $I = [0, 1]$ with the standard subspace topology from $\mathbb{R}$. Define an equivalence relation by $0 \sim 1$, with all other points only equivalent to themselves. The resulting quotient space $I/\sim$ is homeomorphic to the unit circle S^1. If we take a square $I \times I$ and identify $(0, y) \sim (1, y)$ for all y, we get a cylinder. If we instead introduce a twist by identifying $(0, y) \sim (1, 1 - y)$, we obtain the non-orientable Möbius band.

Example 1.15. Further standard quotients of the square $I \times I$ and the disk D^2:

- **Torus:** identify $(0, y) \sim (1, y)$ and $(x, 0) \sim (x, 1)$. First gluing the left and right edges gives a cylinder; then gluing its two boundary circles gives the torus T^2.
- **Klein bottle:** identify $(0, y) \sim (1, y)$ and $(x, 0) \sim (1 - x, 1)$. The cylinder's boundary circles are glued with a reversal of direction, which cannot be done inside $\mathbb{R}^3$ without self-intersection.
- **Sphere:** collapse the entire boundary circle of the closed disk D^2 to a single point. The result $D^2/\partial D^2$ is homeomorphic to S^2 (think of closing a drawstring bag).
- **Projective plane:** identify each point $x \in \partial D^2$ with its antipode $-x$. The result is the real projective plane $\mathbb{R}P^2$.

1.3 Continuous Maps and Homeomorphisms

To compare two topological spaces, we study the structure-preserving maps between them.

Definition 1.16. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces. A function $f : X \to Y$ is **continuous** if for every open set $V \in \mathcal{T}_Y$, the preimage $f^{-1}(V)$ is open in $\mathcal{T}_X$.

If X and Y are metric spaces, this definition is entirely equivalent to the standard ϵ-δ definition of continuity from calculus: points that are close in X map to points that are close in Y.

Example 1.17. Continuity depends on the topologies, not only on the formula.

- Any constant map $f : X \to Y$, $f(x) = y_0$, is continuous: $f^{-1}(V)$ is either X (if $y_0 \in V$) or $\emptyset$.
- Let $\mathbb{R}_{\text{disc}}$ denote $\mathbb{R}$ with the discrete topology. The identity $\text{id} : \mathbb{R}_{\text{disc}} \to \mathbb{R}$ is continuous, since every preimage is open in $\mathbb{R}_{\text{disc}}$. The identity $\text{id} : \mathbb{R} \to \mathbb{R}_{\text{disc}}$ is *not* continuous, since $\{0\}$ is open in $\mathbb{R}_{\text{disc}}$ but not in $\mathbb{R}$.
- The projection $p : X \to X/\sim$ onto a quotient is always continuous, by the very definition of the quotient topology.

The most fundamental equivalence relation in topology is homeomorphism.

Definition 1.18. Let X and Y be topological spaces. A function $f : X \to Y$ is a **homeomorphism** if:

1. f is a bijection.
2. f is continuous.
3. The inverse function f^{-1} is continuous.

If such a map exists, we say X and Y are **homeomorphic**, denoted $X \cong Y$.

Example 1.19. Some homeomorphisms that will be used repeatedly:

- Any two open intervals are homeomorphic: $(a, b) \to (c, d)$, $t \mapsto c + \frac{d-c}{b-a}(t - a)$.
- An open interval is homeomorphic to the whole line: $f : (-1, 1) \to \mathbb{R}$, $f(t) = \dfrac{t}{1 - |t|}$, with continuous inverse $f^{-1}(s) = \dfrac{s}{1 + |s|}$. So boundedness is *not* a topological property.
- **Stereographic projection.** Let $N = (0, 1)$ be the north pole of S^1. The map

$$\phi : S^1 \setminus \{N\} \to \mathbb{R}, \qquad \phi(x, y) = \frac{x}{1 - y},$$

  sends a point P to the intersection of the line through N and P with the x-axis. Its inverse is $\phi^{-1}(s) = \left(\dfrac{2s}{s^2 + 1}, \dfrac{s^2 - 1}{s^2 + 1}\right)$, so $S^1 \setminus \{N\} \cong \mathbb{R}$. The same formula in higher dimensions shows $S^n \setminus \{N\} \cong \mathbb{R}^n$.

Remark 1.20. Condition 3 in the definition of homeomorphism is not automatic. The map $f : [0, 2\pi) \to S^1$, $f(t) = (\cos t, \sin t)$, is a continuous bijection, but its inverse is not continuous at $(1, 0)$: points of S^1 just below $(1, 0)$ are sent to numbers close to 2π, while $(1, 0)$ itself is sent to 0.

To prove two spaces are *not* homeomorphic, one typically finds a topological invariant that differs between them. For instance, the open interval $(0, 1)$ is not homeomorphic to the half-open interval $[0, 1)$. Removing *any* point p from $(0, 1)$ leaves the disconnected space $(0, p) \cup (p, 1)$. However, removing the endpoint 0 from $[0, 1)$ leaves the connected space $(0, 1)$. If $f : [0, 1) \to (0, 1)$ were a homeomorphism, its restriction would be a homeomorphism $[0, 1) \setminus \{0\} \to (0, 1) \setminus \{f(0)\}$ between a connected and a disconnected space, which is impossible.

1.4 Topological Invariants

Topological invariants are properties preserved under homeomorphisms. Two critical invariants are connectedness and compactness.

Definition 1.21. A space X is **connected** if it cannot be written as the disjoint union of two non-empty open sets. It is **path-connected** if for every pair of points $x, y \in X$, there exists a continuous map $\gamma : [0, 1] \to X$ such that $\gamma(0) = x$ and $\gamma(1) = y$.

Every path-connected space is connected, but the converse fails.

Example 1.22.

- Intervals in $\mathbb{R}$, the space $\mathbb{R}^n$, balls, and the spheres S^n $(n \geq 1)$ are path-connected, hence connected.
- $\mathbb{R}\setminus\{0\} = (-\infty, 0)\cup(0, \infty)$ is not connected: it is the disjoint union of two non-empty open sets.
- $\mathbb{Q} \subset \mathbb{R}$ is not connected: $\mathbb{Q} = (\mathbb{Q} \cap (-\infty, \sqrt{2})) \cup (\mathbb{Q} \cap (\sqrt{2}, \infty))$.
- The **topologist's sine curve** $S = \{(0, 0)\}\cup\{(x, \sin(1/x)) \mid 0 < x \leq 1\}$ is connected but not path-connected: no path can reach the origin from the graph, since the graph oscillates infinitely often as $x \to 0$.

The **connected components** of a space are its maximal connected subsets. Their number is a topological invariant; in Week 5 it will reappear as the rank of the zeroth homology group.

Definition 1.23. A collection of open sets $\{U_\alpha\}$ is an **open cover** of X if $X \subset \bigcup U_\alpha$. A space X is **compact** if every open cover has a finite subcover.

Example 1.24. The open interval $(0, 1)$ is not compact. The sets $U_n = (\frac{1}{n}, 1)$, $n \geq 2$, form an open cover of $(0, 1)$, but any finite subcollection $U_{n_1}, \ldots, U_{n_k}$ has union $U_N = (\frac{1}{N}, 1)$ with $N = \max n_i$, which misses the point $\frac{1}{2N}$. Similarly, $\mathbb{R}$ is not compact, as the cover $\{(-n, n)\}_{n\geq 1}$ has no finite subcover. Every finite space is compact.

By the Heine-Borel Theorem, a subspace of $\mathbb{R}^n$ is compact if and only if it is both closed and bounded. Thus $[0, 1]$, S^n, closed balls, and every polyhedron $|K|$ of a finite simplicial complex (defined below) are compact. Two powerful theorems relating continuous functions and compactness are:

Theorem 1.25. *The continuous image of a compact space is compact. If X is compact and $f : X \to Y$ is continuous, then $f(X)$ is compact.*

Theorem 1.26. *Any continuous bijection from a compact space X to a Hausdorff space Y is a homeomorphism.*

Example 1.27. The second theorem gives a quick proof that $I/\sim \cong S^1$, where $0 \sim 1$. The map $f : I \to S^1$, $f(t) = (\cos 2\pi t, \sin 2\pi t)$, is continuous and identifies exactly 0 and 1, so it induces a continuous bijection $\bar{f} : I/\sim \to S^1$. Since $I/\sim$ is compact (the image of the compact space I under the projection) and S^1 is Hausdorff, $\bar{f}$ is a homeomorphism. Compare this with the remark above about $[0, 2\pi)$, which is not compact.

A simple consequence: $[0, 1]$ is not homeomorphic to $(0, 1)$ or to $\mathbb{R}$, since the first is compact and the others are not.

1.5 Simplexes and Polyhedra

As we move towards computational methods, we need a way to describe spaces combinatorially so they can be analyzed by algorithms. The simplest building blocks for this task are simplices (points, line segments, triangles, and their higher-dimensional analogues), which are entirely determined by their vertices.

Definition 1.28. Let $v_0, \ldots, v_k$ be geometrically independent points in $\mathbb{R}^n$ (meaning the vectors $v_1 - v_0, \ldots, v_k - v_0$ are linearly independent). The **k-simplex** σ spanned by these points is the set of all convex combinations:

$$\sigma = \left\{ \sum_{i=0}^{k} t_i v_i \;\middle|\; \sum_{i=0}^{k} t_i = 1 \text{ and } t_i \geq 0 \text{ for all } i \right\}$$

The coefficients t_i in the definition are unique for any point $x \in \sigma$ and are called the **barycentric coordinates** of x. If we restrict one or more of these coefficients to be exactly zero, the resulting subset is spanned by a strict subset of the original vertices. Such a subset is called a **face** of σ. A face spanned by all but one vertex is called a **facet** (which has dimension $k - 1$). The point with all $t_i = \frac{1}{k+1}$ is the **barycenter** of σ.

Example 1.29. Geometric independence matters. The points $(0, 0), (1, 0), (2, 0)$ in $\mathbb{R}^2$ are not geometrically independent, since $(1, 0) - (0, 0)$ and $(2, 0) - (0, 0)$ are parallel; their convex hull is a segment, not a triangle. In $\mathbb{R}^n$ at most $n + 1$ points can be geometrically independent, so $\mathbb{R}^n$ contains k-simplices only for $k \leq n$.

Example 1.30. Let σ be the triangle spanned by $v_0 = (0, 0)$, $v_1 = (4, 0)$, $v_2 = (0, 4)$. The point $x = (1, 2)$ satisfies

$$x = t_0 v_0 + t_1 v_1 + t_2 v_2 = (4t_1, 4t_2),$$

so $t_1 = \frac{1}{4}$, $t_2 = \frac{1}{2}$, and $t_0 = 1 - t_1 - t_2 = \frac{1}{4}$. All three coordinates are positive, so x lies in the interior of σ. The point $(2, 2)$ has $t_0 = 0$ and therefore lies on the facet spanned by v_1, v_2, while $(3, 3)$ has $t_0 = -\frac{1}{2} < 0$ and lies outside σ.

Since a face of a k-simplex corresponds to a non-empty subset of its $k + 1$ vertices, a k-simplex has exactly $\binom{k+1}{j+1}$ faces of dimension j, and $2^{k+1} - 1$ faces in total (including itself). For example, a tetrahedron ($k = 3$) has 4 vertices, 6 edges, 4 triangles and 1 tetrahedron, for a total of $15 = 2^4 - 1$ faces.

To build more complex spaces, we glue simplices together in a highly structured way.

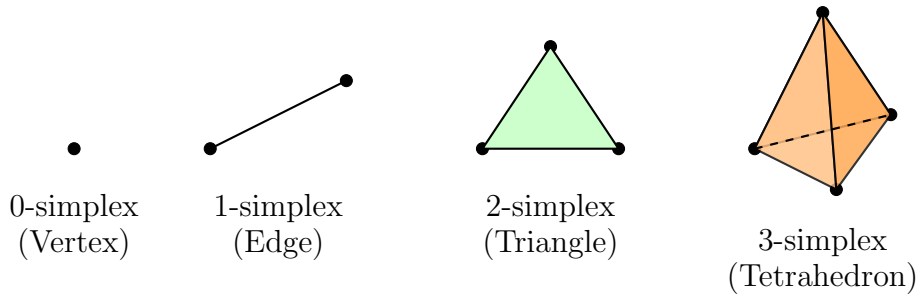


Figure 4: Standard geometric simplices in low dimensions.

Definition 1.31. A **simplicial complex** K in $\mathbb{R}^n$ is a finite collection of simplices such that:

1. Every face of a simplex in K is also in K.
2. The non-empty intersection of any two simplices in K is a face of each.

The first condition ensures that the complex is "closed" under taking boundaries, meaning we do not have a triangle missing its edges. The second condition prevents improper intersections; for instance, a vertex of one triangle cannot rest in the interior of another triangle's edge—it must align exactly with a vertex of the other triangle.

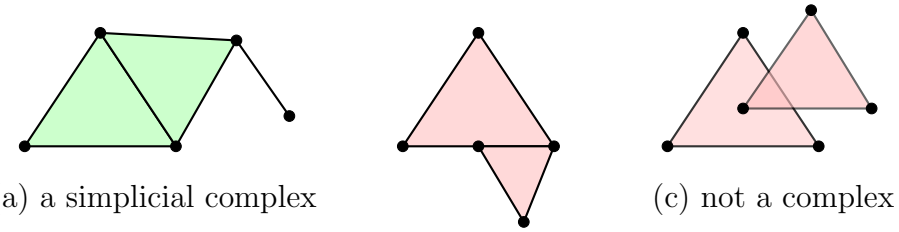


Figure 5: (a) Two triangles sharing an edge, with an extra edge attached at a vertex. (b) The intersection of the two triangles is half of an edge of the upper triangle, which is not a face of it. (c) Two triangles overlapping in a region that is not a face of either.

The **dimension** of the complex K is defined as the maximum dimension of any simplex in K.

Definition 1.32. The union of all simplices in K, denoted $|K|$, is a topological space called the **polyhedron** of K.

It is crucial to distinguish between K and $|K|$. The complex K is a combinatorial object (a set of simplices), which makes it perfect for data structures in algorithms. The polyhedron $|K|$ is a continuous topological space. When a given topological space X is homeomorphic to $|K|$ for some simplicial complex K, we say that K is a **triangulation** of X. We will explore this further when classifying surfaces in Week 2.

Example 1.33.
- Let K consist of a triangle σ together with all of its faces (one triangle, three edges, three vertices). Then $|K| = \sigma$ is a closed disk, so K triangulates D^2.

- Let K' consist only of the three edges and three vertices of σ. Then $|K'|$ is a hollow triangle, homeomorphic to S^1. This is the smallest triangulation of the circle.
- The boundary of a tetrahedron (four triangles, six edges, four vertices) triangulates the sphere S^2.

The same space can have many triangulations: subdividing any edge of K' by a new vertex gives a triangulation of S^1 with four vertices.

2 Week 2: Surfaces and their Triangulations

Having established basic topology, we now specialize to two-dimensional spaces, known as surfaces. The three hours of this week's lectures will cover the definition of surfaces, polygonal presentations, orientability, the connected sum operation, and the complete classification of closed surfaces.

2.1 Manifolds and Surfaces

Definition 2.1. An n-dimensional **manifold** is a second-countable Hausdorff space where every point has an open neighborhood homeomorphic to the open disk in $\mathbb{R}^n$. A **surface** is a 2-dimensional manifold.

Example 2.2.
- $\mathbb{R}^n$ and any open subset of $\mathbb{R}^n$ are n-manifolds.
- The sphere S^n is an n-manifold: every point lies in $S^n \setminus \{N\}$ or $S^n \setminus \{S\}$, each homeomorphic to $\mathbb{R}^n$ by stereographic projection.
- The torus $T^2 = S^1 \times S^1$ is a surface, since a product of two open intervals is an open square, homeomorphic to an open disk.
- The **figure eight** (two circles joined at one point) is not a 1-manifold. A small connected neighborhood of the crossing point minus that point has four components, whereas an open interval minus a point has two.
- Two cones glued at their tips (the double cone $x^2 + y^2 = z^2$) is not a surface: removing the tip from a small connected neighborhood of it leaves two components, whereas an open disk minus a point is connected.

Remark 2.3. The Hausdorff condition is not redundant. The **line with two origins**, obtained from two copies of $\mathbb{R}$ by identifying x in the first copy with x in the second for every $x \neq 0$, is locally homeomorphic to $\mathbb{R}$. However, the two origins cannot be separated by disjoint open sets, so it is not a manifold.

When we allow points to have neighborhoods homeomorphic to the half-disk $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 < 1,\ y \geq 0\}$ (where the boundary of the half-disk corresponds to the boundary of the surface), we get a manifold with boundary. The points with no open-disk neighborhood form the **boundary** ∂S. The standard examples of surfaces with boundary are the closed disk (boundary one circle), the cylinder (boundary two circles) and the Möbius band (boundary one circle). A surface is **closed** if it is compact and has empty boundary.

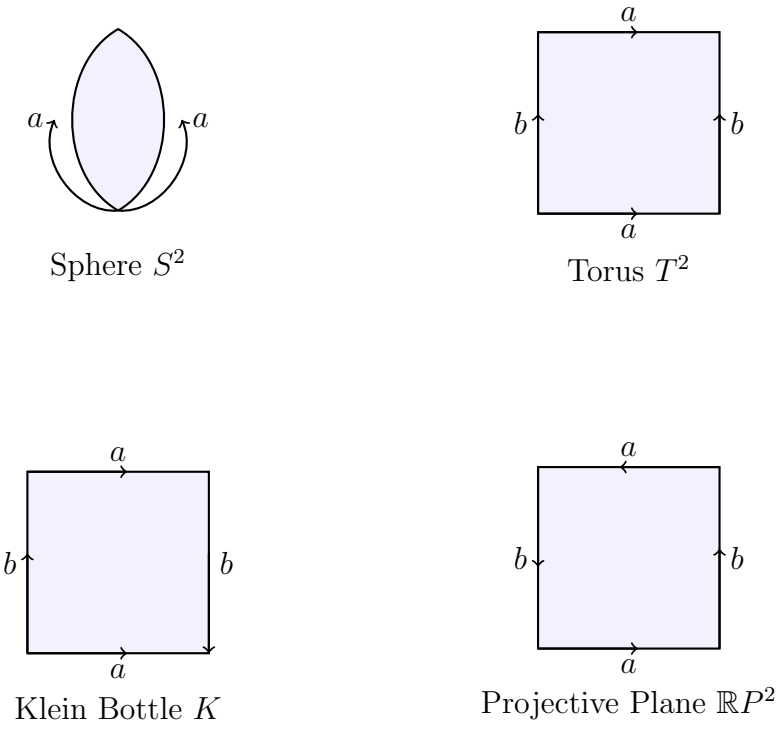


Figure 6: Fundamental polygons for standard closed surfaces. Traversing the boundaries counter-clockwise (starting from the bottom-left corner for the squares) yields the boundary words aa^{-1}, $aba^{-1}b^{-1}$, $ab^{-1}a^{-1}b^{-1}$ and $abab$.

2.2 Polygonal Presentations and Gluing

We can systematically construct compact surfaces by taking a polygon in the plane and identifying its edges in pairs. We represent this gluing using boundary words. We traverse the boundary of the polygon, writing down a letter for each edge. If the arrow points in the direction of traversal, we write a; if it points opposite, we write a^{-1}.

Example 2.4.

- **Sphere S^2:** Take a 2-gon and identify the edges with the word aa^{-1}. The edges fold together like a zipper.
- **Torus T^2:** Take a square and identify opposite edges with the word $aba^{-1}b^{-1}$. Let us trace the vertices: starting at the bottom-left vertex and moving counter-clockwise, the gluing instructions force all four corners of the square to be identified to a single vertex in the quotient space.
- **Klein Bottle K:** A square with the word $ab^{-1}a^{-1}b^{-1}$, as read from the figure. Reversing the arrow on b (that is, renaming b^{-1} as b) gives the equivalent word $aba^{-1}b$.
- **Projective Plane $\mathbb{R}P^2$:** A 2-gon with the word aa, or a square with $abab$. In the square, each point of the boundary is glued to its antipode, exactly as in the description $\mathbb{R}P^2 = D^2/(x \sim -x \text{ on } \partial D^2)$ from Week 1.

Different words can describe the same surface; for instance, the square with $aabb$ is *not* $\mathbb{R}P^2$ but the Klein bottle, as the Euler characteristic computation below shows.

After gluing, a polygonal presentation gives a cell decomposition of the surface: the vertices are the equivalence classes of corners of the polygon, the edges are the letters,

and there is a single face (the interior of the polygon). This lets us compute the Euler characteristic, defined below, directly from the word as

$$\chi = V - E + 1,$$

where E is the number of distinct letters and V is the number of vertex classes.

Example 2.5. We find the vertex classes systematically. Label the corners of the polygon $P_0, P_1, \ldots$ counter-clockwise, so that the i-th edge runs from P_{i-1} to P_i. An edge labelled a goes from its start corner to its end corner in the direction of the arrow; an edge labelled a^{-1} goes the other way. Each pair of edges with the same letter identifies their start corners and their end corners.

The word $aba^{-1}b^{-1}$ (square, corners $P_0, \ldots, P_3$). The edges give $a : P_0 \to P_1$, $b : P_1 \to P_2$, $a : P_3 \to P_2$, $b : P_0 \to P_3$. Gluing the two copies of a gives $P_0 \sim P_3$ and $P_1 \sim P_2$; gluing the two copies of b gives $P_1 \sim P_0$ and $P_2 \sim P_3$. Thus all four corners form one class: $V = 1$, $E = 2$, and $\chi = 1 - 2 + 1 = 0$.

The word aa (2-gon, corners P_0, P_1). The edges give $a : P_0 \to P_1$ and $a : P_1 \to P_0$, so $P_0 \sim P_1$. Hence $V = 1$, $E = 1$, $\chi = 1$.

The word $abab$ (square). The edges give $a : P_0 \to P_1$, $b : P_1 \to P_2$, $a : P_2 \to P_3$, $b : P_3 \to P_0$. Gluing a gives $P_0 \sim P_2$, $P_1 \sim P_3$, and gluing b gives the same two identifications. Hence $V = 2$, $E = 2$, $\chi = 1$, as for the 2-gon aa.

The word $aabb$ (square). The edges give $a : P_0 \to P_1$, $a : P_1 \to P_2$, $b : P_2 \to P_3$, $b : P_3 \to P_0$, so $P_0 \sim P_1 \sim P_2$ and $P_2 \sim P_3 \sim P_0$. Hence $V = 1$, $E = 2$, $\chi = 0$. This surface is non-orientable with $\chi = 0$; by the classification below it is the Klein bottle $\mathbb{R}P^2 \# \mathbb{R}P^2$.

The word aa^{-1} (2-gon). The edges give $a : P_0 \to P_1$ and $a : P_0 \to P_1$ (the second one is traversed backwards), so no corners are identified. Hence $V = 2$, $E = 1$, $\chi = 2$.

The word $a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1}$ ($4g$-gon). As in the torus case all corners are identified to one vertex, so $V = 1$, $E = 2g$ and $\chi = 2 - 2g$.

The algorithmic classification of surfaces largely relies on manipulating these boundary words through "cut and paste" operations (e.g., cutting a polygon along a diagonal and re-gluing along a different pair of identified edges) to bring any word into a standard canonical form.

2.3 Orientability and Connected Sums

A surface is **non-orientable** if it contains a subspace homeomorphic to the Möbius band; otherwise, it is **orientable**. For instance, the sphere and torus are orientable, while the Klein bottle and projective plane are non-orientable. A surface given by a single polygon with edges identified in pairs is non-orientable if and only if some letter appears twice with the same sign (e.g., a and a in $abab$).

Example 2.6. In the word $abab$ for $\mathbb{R}P^2$, consider a thin vertical strip in the middle of the square, running from the bottom edge a to the top edge a. The bottom arrow points right and the top arrow points left, so the left side of the strip at the bottom is glued to the right side of the strip at the top: the strip is glued with a half-twist and becomes a Möbius band in $\mathbb{R}P^2$. In the torus word $aba^{-1}b^{-1}$ both arrows point the same way, the same strip is glued without a twist, and it becomes a cylinder.

To generate more complex surfaces, we use the connected sum operation.

Definition 2.7. Let S_1 and S_2 be surfaces. The **connected sum** $S_1 \# S_2$ is formed by cutting an open disk out of each surface and gluing the resulting boundary circles together.

For example, the connected sum of two tori $T^2 \# T^2$ is a genus-2 surface (a sphere with two handles). The sphere is the identity for this operation: $S \# S^2 \cong S$ for every surface S, because removing a disk from S^2 leaves a disk, which simply fills the hole in S. In terms of words, the connected sum is obtained by concatenation: if S_1 and S_2 have words w_1 and w_2 (with disjoint letters), then $S_1 \# S_2$ has word $w_1 w_2$. For example, $T^2 \# T^2$ has word $aba^{-1}b^{-1}cdc^{-1}d^{-1}$, and $\mathbb{R}P^2 \# \mathbb{R}P^2$ has word $aabb$, which we identified above as the Klein bottle.